SOLUTION

1. CONSTRAINT-BASED PROBLEM SOLVING USING BACKTRACKING SEARCH

Step 1: Assign D1

- Try **Morning** ✓
- Constraint satisfied (D1 cannot work Night).

Assignment: D1 → Morning

Step 2: Assign D2

- Morning ✗ (already assigned)
- Afternoon ✓

Assignment: D2 → Afternoon

Step 3: Assign D3

- Morning ✗ (D3 cannot work Morning)
- Afternoon ✗ (already assigned)
- Night ✓

Assignment: D3 → Night

Constraint Checking

CONSTRAINT	RESULT
D1 cannot work Night	✓ Satisfied
D2 before D3	✓ Afternoon before Night
One doctor per shift	✓ Satisfied
D3 cannot work Morning	✓ Satisfied
One shift for each doctor	✓ Satisfied

Final Assignment

DOCTOR	SHIFT
D1	Morning
D2	Afternoon
D3	Night

Solution: All constraints are satisfied. Backtracking finds the valid assignment without needing to backtrack further.

2 ROBOT PATH FINDING USING SEARCH ALGORITHM

Assume the following grid:

S □ □ X □

□ X □ X □

□ □ □ □ □

X X □ X □

□ □ □ □ G

S = Start, G = Goal, X = Obstacle

Step 1

Start at **S**.

Queue = [S]

Step 2

Expand S.

Queue = [(0,1), (1,0)]

Step 3

Expand the next node.

Blocked cells are ignored.

Queue = [(1,0), (0,2)]

Step 4

Continue expanding nodes.

Visit only unvisited safe cells.

Step 5

Eventually Goal (G) is reached.

Optimal Path

S → (0,1) → (0,2) → (1,2)

→ (2,2) → (2,3) → (2,4)

→ (3,4) → G

Total Cost

Number of moves = **8**

Cost per move = **1**

Total Cost = 8

3.AUTONOMOUS RESCUE ROBOT

(a) Constraint Analysis

- Robot moves only up, down, left, and right.
- Obstacles block movement.
- Robot observes only nearby cells.
- Every move costs 1.
- Risk zones add extra cost.
- Robot updates its map while moving.

These constraints force the robot to choose the safest and least-cost route.

(b) Solution Using Uniform Cost Search (UCS)

Steps

1. Start from **S**.
2. Add S to the priority queue with cost 0.
3. Expand the node with the lowest cost.
4. Observe neighboring cells.
5. Ignore blocked cells.
6. Calculate movement and risk costs.
7. Insert new nodes into the priority queue.
8. Repeat until Goal **G** is reached.
9. Trace back the minimum-cost path.

(c) AI Concepts

Intelligent Agent

The robot senses the environment, makes decisions, and performs actions autonomously.

Rationality

It always chooses the action with the lowest total cost while avoiding risky areas.

Environment Type

- Partially observable
- Dynamic
- Sequential
- Discrete
- Unknown

(d) Justification

Uniform Cost Search is appropriate because:

- It always finds the minimum-cost path.
- It considers both movement and risk costs.

- It avoids expensive routes.
- It guarantees an optimal solution.
- It is suitable for disaster rescue where safety is important.

Conclusion

The rescue robot successfully reaches the survivor by avoiding obstacles, minimizing cost, and updating its knowledge of the environment. Uniform Cost Search provides an optimal and reliable solution under the given constraints.